Physiology of Testicular Function

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Block: Endo/ Uro/ Repro

Thread: Physiology

Conflict of Interest: Dr. Haines has no conflicts of interest to disclose.

Learning Objectives

By exam time, you should be able to do the following:

1. **Explain** the role of gene regulatory proteins (SRY/TDF), Leydig cells, Sertoli cells, and their hormonal products in testicular development and sexual differentiation.
2. **Predict** the developmental consequences of impaired androgen production or signaling during sexual differentiation.
3. **Describe** the regulation of testicular function by the hypothalamic-pituitary-gonadal (HPG) axis, including the actions and feedback mechanisms of GnRH, LH, FSH, testosterone, estradiol, and inhibin.
4. **Compare** the actions of testosterone, dihydrotestosterone (DHT), and estradiol in reproductive development, puberty, and adult reproductive function.
5. **Explain** the stages of spermatogenesis and the roles of Sertoli cells, Leydig cells, and reproductive hormones in regulating sperm production.
6. **Explain** the neural and vascular mechanisms underlying erection and ejaculation.

Study Questions

1. How does SRY drive the development of embryonic testis and initiate testosterone production?

2. What are the main genotype and phenotype variations in the testicular reproductive system?

3. What are the major stages in testosterone production during the life cycle?

4. How does the hypothalamic-pituitary axis regulate testosterone production?

5. Identify the major physiological actions of testosterone, DHT and estradiol in the XY-individual.

6. Describe the physiology of spermatogenesis.

7. Describe the basics of communication between Leydig cells and Sertoli cells.

A. Introduction

1. Human reproduction involves sperm production by the person with testes, which requires normal development of genital organs, production of androgens, proliferation of the germ cell and differentiation into mature sperm.

2. A testicular fertility problem is present in nearly half of all infertile couples. To solve infertility problems, we must understand the mechanism underlying the production and maturation of sperm.

3. A gene region on the Y chromosome initiates the orchestrated regulation of the development, maturation and functions of the testicular reproductive system, which are driven through endocrine signaling, feedback and crosstalk by gonadotropins and androgens.

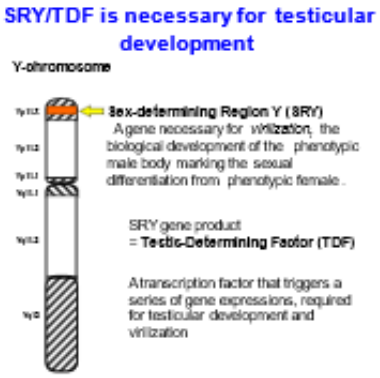
4. Genetic disorders in the system, such as impaired androgen production or sensitivity, result in infertility and developmental disorders. In addition, because of their anabolic effects, abuse of androgens is common among healthy children and adults of both sexes.

B. Genetic program in testicular genital development

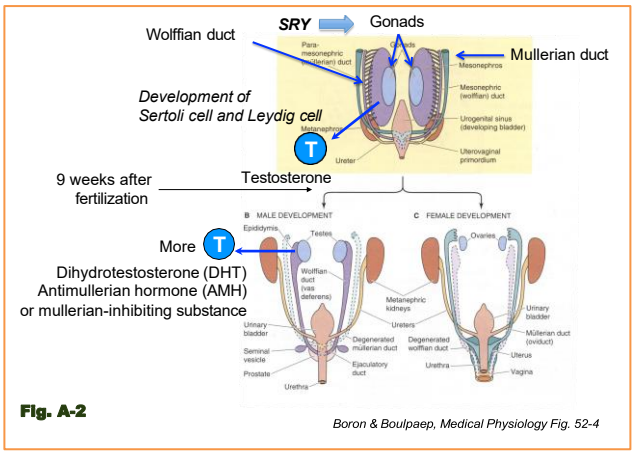
1. SRY (sex-determining region Y) / TDF (testis-determining factor) is necessary for virilization.

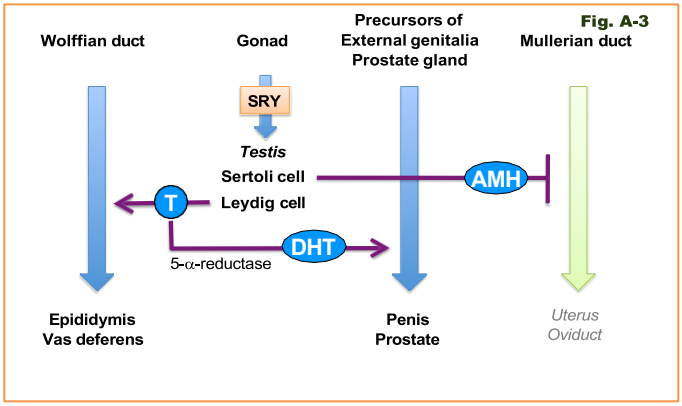
2. Human chromosomes consist of a pair of sex chromosomes plus 22 pairs of autosomal chromosomes. The sex chromosomes, X and Y, carry the genetic information that determines reproductive organ development. **(Fig. A-1)**. A combination of two X chromosomes derived from each gamete, results in XX genotype, leading the development of ovaries. On the other hand, the XY genotype, an X from the oocyte and a Y from sperm, leads to the development of testis.

3. A small region on the Y chromosome has been identified as SRY, sex-determining region Y. An intact structure of SRY and the function of the gene product, named TDF, testis- determining factor (a transcription factor), is necessary for the biological processes of testicular development and virilization.



4. SRY drives the development of embryonic testis, which initiates testosterone production: As shown in **Fig. A-2**, the undifferentiated gonad in the early embryo is sexually indistinguishable, consisting of both Mullerian ducts and Wolffian ducts, which are to be oviduct and vas deferens, respectively (**Fig. A-2, top**).





5. **Fig. A-3** summarizes the development of the genital tract. The activation of the genes encoded in SRY in the embryonic gonad induces the differentiation of Sertoli cells, followed by Leydig cells. Sertoli cells start producing anti-Mullerian hormone (AMH), a substance forcing Mullerian duct regression. Leydig cells produce testosterone (T) that induces the growth and differentiation of the Wolffian ducts, and the testosterone-derived DHT triggers growth of the penis and prostate.

C. Genotype vs phenotype in testicular reproductive system

1. Genetic analysis of the developmental differences in the testicular reproductive system shows the role of SRY, testosterone, and AMH in sexual differentiation. XO genotype, lacking an entire Y chromosome (also known as Turner's syndrome) results in the partial development of the gonads and Mullerian ducts and complete regression of Wolffian duct, due to a lack of SRY and Leydig and Sertoli cells. Unlike XO genotype, SRY deficiency results in immature testis, suggesting a partial development of testis due to the rest of the Y chromosome. People with XO or SRY deficiency have external genitalia aligned with the biologic female.

2. A deficiency in testosterone signal, such as the distortion in testosterone synthesis or receptor, causes regressed Wolffian ducts and partial development of the penis and prostate. Deficiency in AMH leads to Wolffian development, and partial development of Mullerian duct through activity on the X chromosome.

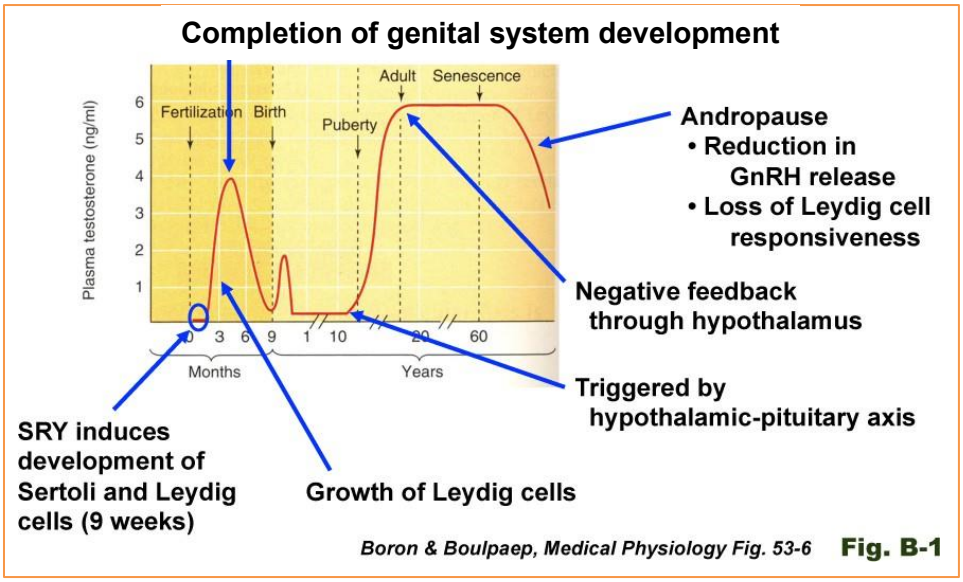
3. Individuals who carry a 47_{th} chromosome, XXY genotype, also known as Klinefelter's syndrome, typical possess a vas deferens and penis, with partial deficiency in testis function. The XX genotype with a translocated SRY causes the phenotype similar to XXY, although the individuals are generally infertile.

Genotype	*Gonad*	*Mullerian duct*	*Wolffian duct*	*Phenotype*
XY, Normal	**testis**	**Regressed**	**Developed**	**M**
XX, Normal	**ovary**	**Developed**	**Regressed**	**F**
XO, Turner's syndrome	**streak**	**Developed**	**Regressed**	**F**
XY w/ Deficiency in SRY	**testis***	**Developed**	**Regressed**	**F**
XY w/ Deficiency in testosterone signal	**testis**	**Regressed**	**Regressed****	**M/F**
XY w/ Deficiency in AMH signal	**testis**	**Developed**	**Developed**	**M**
XXY, Klinefelter's syndrome	**testis***	**Regressed**	**Developed**	**M**
XX w/ SRY translocation	**testis***	**Regressed**	**Developed**	**M**

** Immature testis development, mostly sterile.*
*** Depending on the degree of deficiency, partial Wolffian Duct development may occur.*

D. Androgen physiology

1. Developmental changes in plasma testosterone level: Testosterone and related androgens play the dominant role in the development of genital organs and sperm production. **Fig. B-1** shows a typical profile of plasma testosterone during the lifespan of a an XY individual.



2. 1st surge in testosterone production:
   a. Around 9 weeks after fertilization.
   b. Dependent on the development and growth of embryonic Leydig cells.
   c. This testosterone production is necessary for the genital development in the fetus.

3. Dormant stage:
   a. After completion of anatomical development.
   b. Counts of Sertoli cells and Leydig cells in embryonic testis are reduced.
   c. Continues until puberty starts.
   d. Excess testosterone in the pre-pubertal XY individual causes the appearance of puberty.

4. Saturation stage:
   a. The level of testosterone is maintained at approximately 6 ng/mL.
   b. The maintenance of the plasma level is due to negative feedback through the hypothalamic-pituitary axis.
   c. This level of testosterone is necessary for normal spermatogenesis.

5. Andropause: Age-related reduction in testosterone level
   a. It is caused by reduction in GnRH release and loss of Leydig cell responsiveness. Diminished libido, erectile capacity, and sperm production.
   b. Associated with fatigue, depression, decreased body mass and bone mineral density.

6. Puberty : the physiologic transition between childhood and adulthood, and involves the development of fertility, secondary sexual characteristics and the pubertal growth spurt.

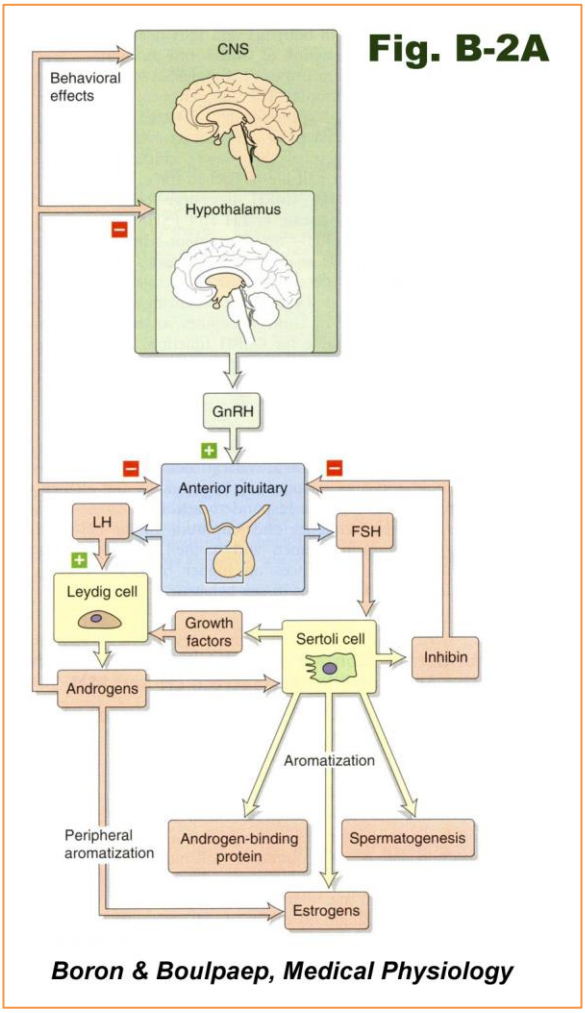
E. Hypothalamic-pituitary axis regulates testosterone production

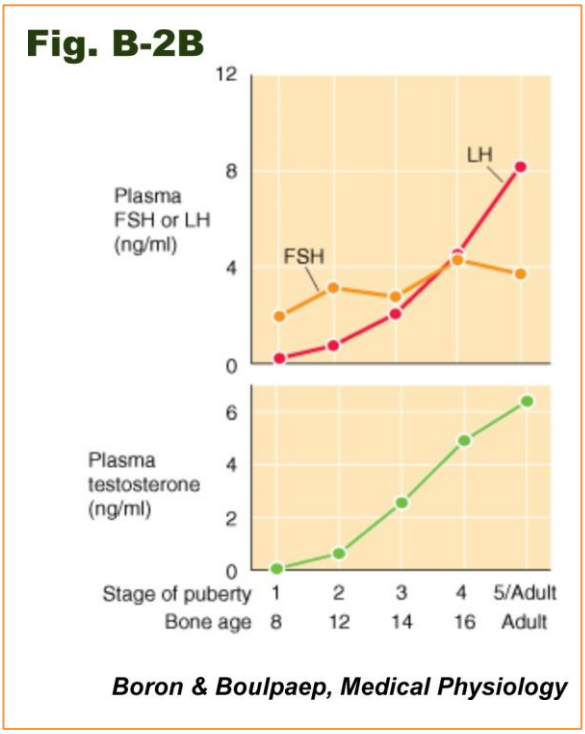
1. **Fig B-2A** shows overview of the endocrine regulation of testis functions. During the dormant stage, the GnRH pulse generator is suppressed. The increase in the pulsatile secretion of GnRH from the hypothalamus induces the release of LH and FSH from the anterior pituitary **(Fig. B-2A)**.

2. The mechanism initiating pulsatile GnRH release remains unclear. Current evidence suggests the involvement of an increased adrenal production of DHEA and leptin secreted from adipose tissue. Pulsatile GnRH release is essential to activation of the pituitary, because a continuous administration of GnRH reduces LH release.

3. LH stimulates Leydig cells that synthesize and release testosterone **(Fig. B-2A and B-2B)**. Testosterone and the related androgens induce a number of physiological actions required for pubertal development and growth, physically and mentally.

4. The high levels of testosterone and estradiol converted from testosterone suppress the activities of the hypothalamus and pituitary, respectively, resulting in negative feedback regulation **(Fig. B-2A)**.





F. Communication between Leydig cells and Sertoli cells.

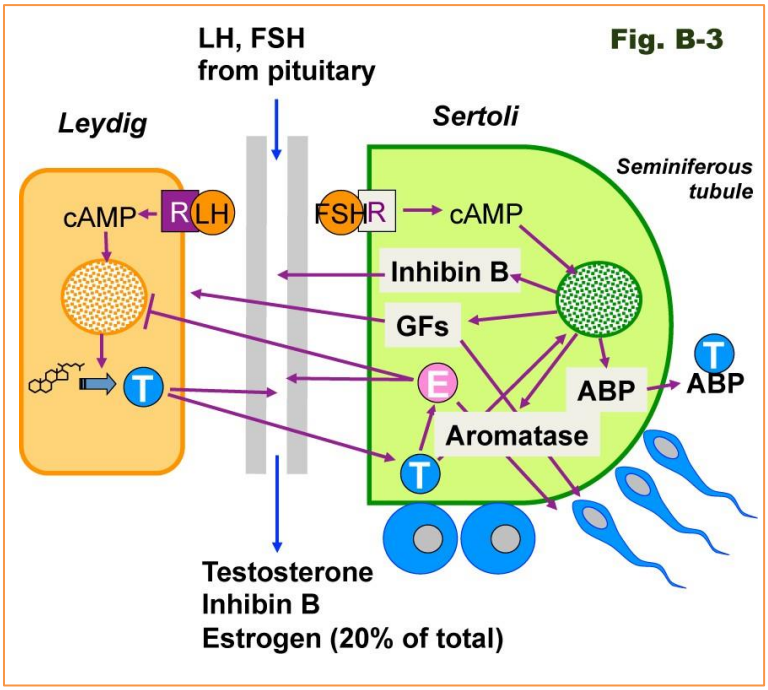
1. The gonadotropins, LH and FSH, released from the pituitary, travel through the general circulation and directly activate Leydig cells and Sertoli cells, respectively. In both cells, the gonadotropin stimulus increases cAMP levels **(Fig. B-3)**.

2. In Leydig cells, cAMP signals triggers the expression of a subset of genes

required for testosterone synthesis from the precursor cholesterol.

3. In Sertoli cells, cAMP elevation triggers the gene activation that produces 1) inhibin B, 2) growth factors (GFs), 3) aromatase, and 4) androgen-binding protein (ABP).

4. **Fig. B-3** summarizes the communication among Leydig cells, Sertoli cells and the germ cells in testis. FSH-induced production of growth factors (GFs) in Sertoli cells stimulates the growth and differentiation of Leydig cells and spermatogenesis. Testosterone produced at Leydig cells permeates into Sertoli cells.



5. Testosterone facilitates FSH signaling in Sertoli cells, as well as binds to ABP produced by Sertoli cells, which is critical for spermatogenesis. Testosterone is also converted into estradiol by aromatase in Sertoli cells. Estradiol plays a positive role in sperm maturation, but a negative role in Leydig cell functions. An increase in the germ cells is known to trigger Inhibin B production, although the mechanisms are unknown. Inhibin B is released into plasma and suppresses the pituitary (see **Fig. C-3**). This inhibin B-induced suppression of the pituitary is a negative feedback pathway regulating spermatogenesis (further *discussed in Fig. C-2*).

G. Distribution of androgens

1. Testosterone is highly cell-permeable **(Fig. B-4)**. Leydig cells produce 95 % of total testosterone, and are the only cells synthesizing testosterone from cholesterol. Testosterone bound to ABP does not permeate the testis-blood barrier formed by Sertoli cells, and is localized at the lumen of seminiferous tubules. Testosterone is also trapped by circulating serum albumin and globulins and transferred into peripheral organs. In addition, 5 % of the total testosterone is converted *de novo* at peripheral organs, such as adipose, brain, muscle, skin and adrenal cortex, from circulating dehydroepiandrosterone (DHEA) and androstenedione. Testosterone is metabolized primarily in the liver into water-soluble forms, prior to excretion in the urine and the feces.

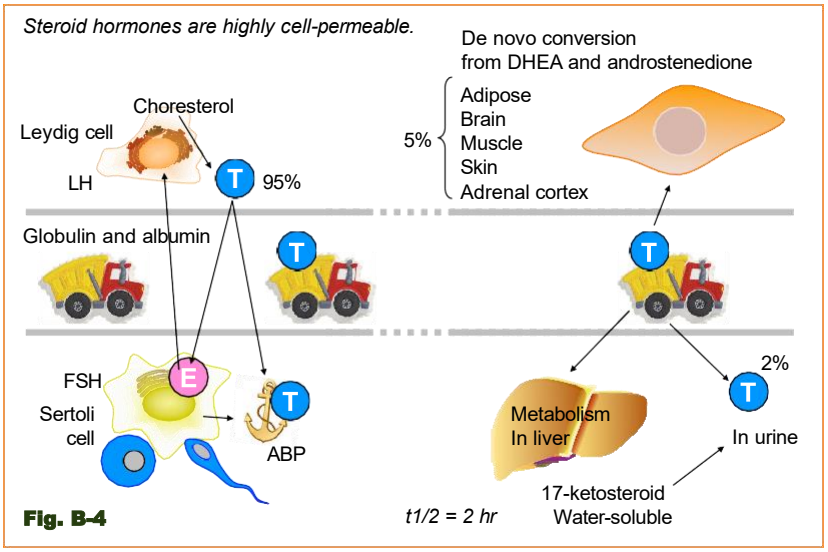
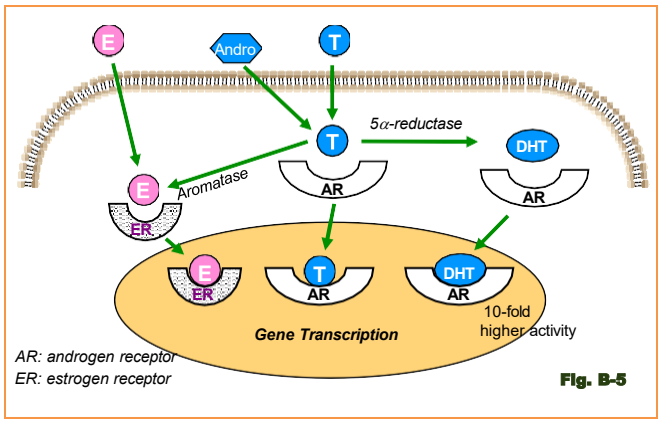


Fig. B-4

H. Cell signaling of androgens at targets

1. **Fig. B-5** illustrates the mechanism by which testosterone controls gene expression. In addition to plasma testosterone, the precursor androgens, such as DHEA and androstenedione ("Andro" in the figure) are converted to testosterone at the target cells. Testosterone and DHT converted by 5D-reductase directly binds to AR.

2. Compared to testosterone, DHT more potently binds to AR and produces over 50-fold higher activity in gene transcription. Estradiol converted from testosterone activates ER to control gene expression, in addition to the circulating estrogen.



I. Physiological actions of testosterone, DHT and estradiol in the XY individual

1. Because 5D-reductase expression is restricted to specific cell types, DHT primarily functions at specific loci. It is very important to understand the androgenic and anabolic effects of each sex hormone.

2. <u>Testosterone</u>

   Development of **internal** genital tract (Wolffian duct-derived structure) epididymis, vas deferens, seminal vesicles

   a. Pubertal growth of penis and seminal vesicle

   b. Stimulation of spermatogenesis

   c. Libido

   d. Pubertal growth of larynx and deepening of voice

   e. Pubertal growth spurt: anabolic effects on muscle, bones, and erythropoiesis Inhibition of breast development

   f. Feedback inhibition of GnRH, LH, and FSH release

3. <u>DHT</u> <<< *5α-reductase*

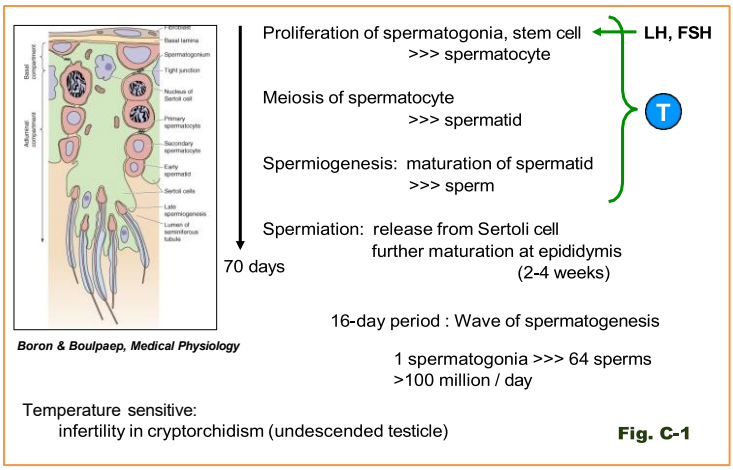
Development of prostate, penis, and scrotum Development of pubic and underarm hair, “male pattern” balding, Activity of sebaceous glands (acne)

4. **Estradiol** <<< *Aromatase*
   a. Epiphyseal growth plate closure, prevention of osteoporosis (Ca^{2+} retention)
   b. Feedback inhibition of GnRH secretion
   c. Spermatogenesis

J. Spermatogenesis, erection, and ejaculation response

1. **Physiology of spermatogenesis:** Maturation of the stem cell into sperm occurs around Sertoli cells that nurture the germ cells, through the production of growth factors and specific substances required for the proliferation and differentiation in spermatogenesis. The process of spermatogenesis is summarized in **Fig. C-1**.

2. The maturation process from the stem cells (spermatogonia) into mature sperm takes approximately 70 days. One spermatogonium produces 64 sperms. A synchronized maturation of a group of spermatogonia starts every 16 days, which is called "wave of spermatogenesis". An average production rate of sperm by an adult is over 100 million sperm per day.



3. Sperm cells will not mature above core body temperature. Therefore, testicular descent is a necessary process for fertility. A condition called *cryptorchidism*, in which testes fail to descend into the scrotum, causes infertility.

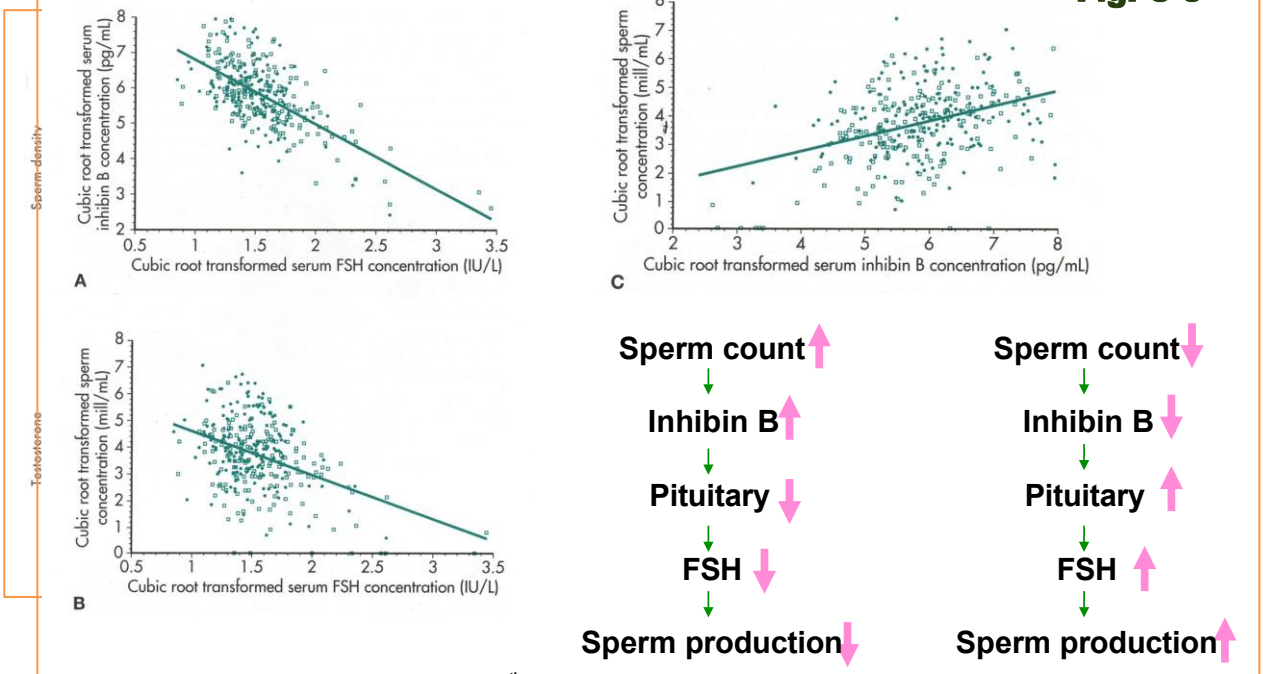
4. A pathological reduction in testosterone level or deficiency in AR causes both loss of sperm count and impaired maturation of sperm, indicating that *testosterone is necessary for both the proliferation of spermatogonia and maturation of sperm.*

K. Roles of testosterone, LH, and FSH in spermatogenesis

1. **Fig. C-2** shows the results of a clinical study of human spermatogenesis. Volunteer XY individuals with normal-range testosterone level were administered synthetic testosterone, which caused a reduction in plasma LH and FSH, due to the negative feedback to pituitary. In parallel, the sperm count was *reduced* at higher level of total testosterone, but the morphology and motility of the sperm were normal.

2. The data support the concept that testosterone is *not sufficient* for the proliferation of stem cells, although it is *necessary* for proliferation and maturation (see **Fig. C-1**). When FSH or LH was individually given to near the normal level, sperm count was restored without a significant change in plasma testosterone level. *These results show that LH and FSH are directly involved in the proliferation of spermatogonia through the activation of Sertoli cells.* The importance of FSH and LH in maintaining sperm count is evident, because administration of FSH and hCG (as an LH substitute) has been used for the treatment of testicular infertility. Normally, FSH deficiency does not cause functional defects of sperm.

3. **Inhibin B levels correlate with total sperm count: Fig. C-3** shows the relationship of the sperm count, serum FSH and inhibin B in a steady state. As discussed above, serum FSH and inhibin B have a reciprocal relationship with each other, due to the negative feedback regulation of pituitary by inhibin B **(Fig. C-3, panel A)**. The sperm count is related to inhibin B level **(panel C)**, but inversely related to FSH **(panel B)**. The data suggest that the germ cells send signals to Sertoli cells to produce inhibin B that suppresses basal FSH production from the pituitary. When the sperm count falls, the reduction in inhibin B results in an increase in FSH, shifting toward spermatogenesis. Serum inhibin B level is indicative of spermatogenesis.



L Sexual functioning

1. Erection and ejaculation response are governed by a sequential activation of parasympathetic and sympathetic nerves, and a coordinated contraction of smooth muscles and striated muscles. Nerve damage due to trauma and surgical processes, such as prostatectomy, may result in erectile dysfunction.

2. **Erection**

   Parasympathetic neuron signal (Ach) activates vascular endothelium to produce nitric oxide (NO) in the corporeal vasculature. NO induces vascular smooth muscle relaxation, increases blood flow, and blood fill in corpus cavernosum.

3. **Emission**

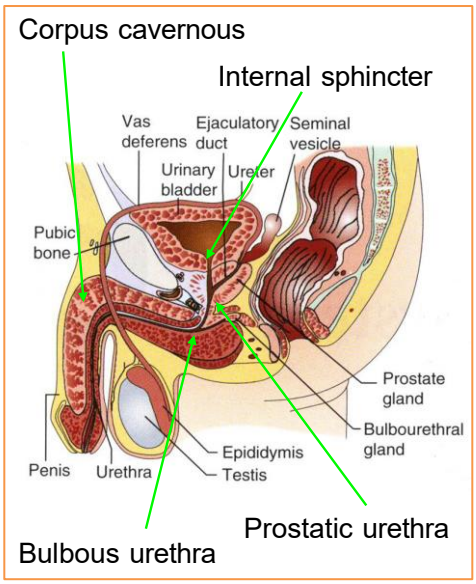
   Sympathetic neurons induce peristaltic motions of vas deferens, seminal vesicles and prostatic smooth muscles, and transfers seminal fluid to the prostatic urethra. Contraction of internal sphincter of bladder blocks ejaculation of sperm into bladder.

4. **Ejaculation**

   Entry of semen to bulbous urethra triggers a reflex reaction, which induces the contraction of the perineal striated muscles to push semen out. A typical ejaculate contains 200-400 million sperm in a volume of 2-4 mL. The sperm movement reaches a velocity of 2-3 mm/min after ejaculation.

5. Detumescence

Sympathetic nerves induce the contraction of corporeal vasculature.



Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. How does SRY drive the development of embryonic testis and initiate testosterone production?

   SRY drives the development of embryonic testis, which initiates testosterone production. The activation of the genes encoded in SRY in the embryonic gonad induces the differentiation of Sertoli cells, followed by Leydig cells. Sertoli cells start producing anti- Mullerian hormone (AMH), a substance forcing Mullerian duct regression. Leydig cells produce testosterone (T) that induces the growth and differentiation of the Wolffian ducts, and the testosterone-derived DHT triggers growth of the penis and prostate.

2. What are the main genotype and phenotype variations in the testicular reproductive system?

Genotype	*Gonad*	*Mullerian duct*	*Wolffian duct*	*Phenotype*
XY, Normal	**testis**	**Regressed**	**Developed**	**M**
XX, Normal	**ovary**	**Developed**	**Regressed**	**F**
XO, Turner's syndrome	**streak**	**Developed**	**Regressed**	**F**
XY w/ Deficiency in SRY	**testis***	**Developed**	**Regressed**	**F**
XY w/ Deficiency in testosterone signal	**testis**	**Regressed**	**Regressed****	**M/F**
XY w/ Deficiency in AMH signal	**testis**	**Developed**	**Developed**	**M**
XXY, Klinefelter's syndrome	**testis***	**Regressed**	**Developed**	**M**
XX w/ SRY translocation	**testis***	**Regressed**	**Developed**	**M**

**** Immature testis development, mostly sterile.***
***** Depending on the degree of deficiency, partial Wolffian Duct development may occur.***

***** Depending on the degree of deficiency, a partial development of male genitalia may occur.***

3. What are the major stages in testosterone production during the life cycle?

   First surge in testosterone production, dormant stage, puberty, saturation stage, andropause.

4. How does the hypothalamic-pituitary axis regulate testosterone production?

   An increase in the pulsatile secretion of GnRH from the hypothalamus induces the release of LH and FSH from the anterior pituitary.

   Pulsatile GnRH release is essential to activation of the pituitary, because a continuous administration of GnRH reduces LH release.

   LH stimulates Leydig cells that synthesize and release testosterone. Testosterone and the related androgens induce a number of physiological actions required for pubertal development and growth.

   The high levels of testosterone and estradiol converted from testosterone suppress the activities of the hypothalamus and pituitary, respectively, resulting in negative feedback regulation

5. Identify the major physiological actions of testosterone, DHT and estradiol in the XY individual.

   Testosterone

   Development of **internal** genital tract (Wolffian duct-derived structure) epididymis, vas deferens, seminal vesicles

   Pubertal growth of penis and seminal vesicles Stimulation of spermatogenesis

   Libido

   Pubertal growth of larynx and deepening of voice

   Pubertal growth spurt: anabolic effects on muscle, bones, and erythropoiesis Inhibition of breast development

   Feedback inhibition of GnRH, LH, and FSH release

DHT

Development of prostate, penis, and scrotum.

Development of pubic and underarm hair, "male pattern" balding

Activity of sebaceous glands (acne)

Estradiol

Epiphyseal growth plate closure, prevention of osteoporosis (Ca^{2+} retention)
Feedback inhibition of GnRH secretion

Spermatogenesis

6. Describe the physiology of spermatogenesis.

Maturation of the stem cell into sperm occurs around Sertoli cells that nurture the germ cells, through the production of growth factors and specific substances required for the proliferation and differentiation in spermatogenesis.

Testosterone is *not sufficient* for the proliferation of stem cells, although it is *necessary* for proliferation and maturation. LH and FSH are directly involved in the proliferation of spermatogonia through the activation of Sertoli cells. Inhibin B levels correlate with total sperm count.

7. Describe the basics of communication between Leydig cells and Sertoli cells.

FSH-induced production of growth factors (GFs) in Sertoli cells stimulates the growth and differentiation of Leydig cells and spermatogenesis. Testosterone produced at Leydig cells permeates into Sertoli cells.